
Lecture 13: Line Emission Mechanisms

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Maser

Maser: Microwave Amplification by Stimulated Emission of Radiation $\Rightarrow$ Non-thermal!!

Theory:

NH₃ (Gordon et al. 1954)

1964 Nobel Prize Given to Basov, Prokhorov, Townes

Cosmic Discovery:

Such emission is observed from molecules such as hydroxyl radicals (OH, Weaver et al. 1965), water (H₂O), methanol (CH₃OH), formaldehyde (H₂CO), and silicon monoxide (SiO).

Properties:

- compact (eg., 10^{14} cm)
- high brightness temperature (e.g., 10^{10} - 10^{12} K)
- Narrow line width (e.g, 0.2–0.4 km/s)
- significant polarization
- time variable

Maser - nonthermal

The maser phenomenon is very natural in interstellar space, since there are usually **deviations from LTE**. Masers arise when the population of the upper energy level, n_u , is more than the factor $(g_u/g_l)n_l$, i.e., **population inversion**. This is formerly represented by a **negative T_{ex}** in the expression

$$\frac{n_u}{n_l} \equiv \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right)$$

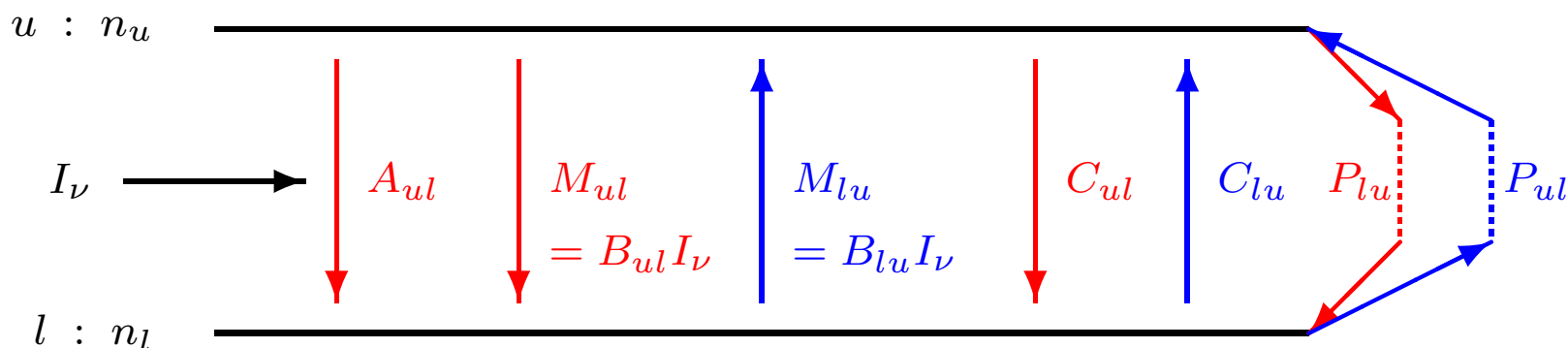
i.e,

$$\Rightarrow T_{\text{ex}} < 0 \quad \text{if} \quad n_u > n_l \frac{g_u}{g_l}$$

For this situation, the transition connecting n_u with n_l can **amplify a background source of radiation**, if one is present. Note that if there is no background continuum and if the line optical depth is small, no masering line will be produced.

Maser Mechanism

Let us consider a cloud of molecules with three energy levels, l , u , and 3. For simplicity, $g_u = g_l$, so that $B_{ul} = B_{lu} = B$.



Here P_{lu} and P_{ul} are the **pumping rates e.g., by IR radiation** from levels l and u to a third level and then back down to levels u and l , respectively. **Such a transfer to a third level is essential to produce population inversion (i.e, negative T_{ex}).**

The radiation transfer eq. corresponding to the transition $l \rightleftharpoons u$ with the frequency ν_0 is then

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} [(n_u - n_l)BI_\nu + n_u A] \phi(\nu)$$

For a steady state,

$$n_u(A_{ul} + M_{ul} + C_{ul} + P_{ul}) = n_l(M_{lu} + C_{lu} + P_{lu})$$

Maser Mechanism

For a steady state,

$$n_u(A_{ul} + M_{ul} + C_{ul} + P_{ul}) = n_l(M_{lu} + C_{lu} + P_{lu})$$

For simplification, we can assume

$$C_{ul} \approx C_{lu} \approx C$$

$$M_{ul} \approx M_{lu} \approx M$$

$$A \ll C, A \ll M$$

then

$$\frac{n_u}{n_l} = \frac{P_{lu} + M + C}{P_{ul} + M + C}$$

If $C = M = 0$, the population inversion would be

$$(n_u - n_l)|_{M=C=0} = n \frac{P_{lu} - P_{ul}}{P} \equiv \Delta n_0$$

with

$$n = n_u + n_l, \Delta n = n_u - n_l, P = P_{lu} + P_{ul}$$

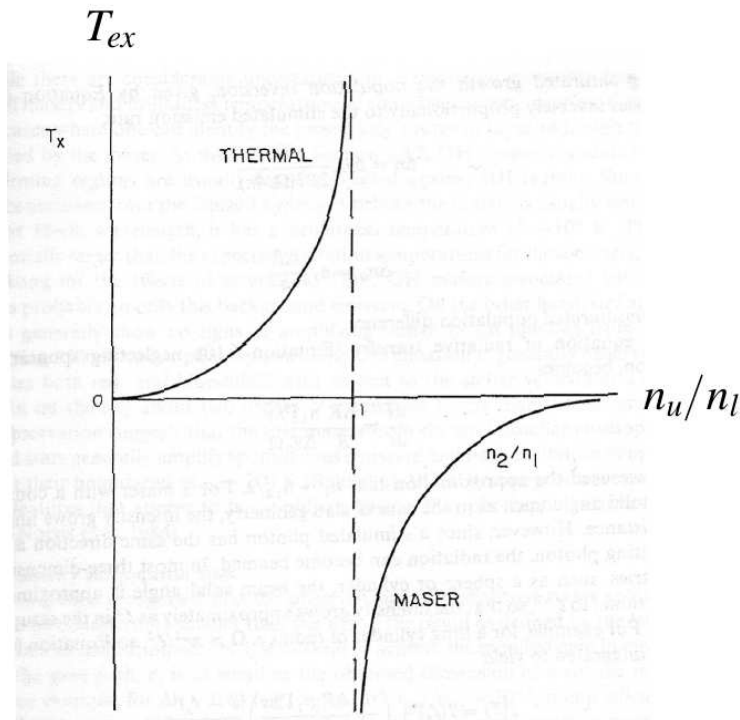
Maser Mechanism

In a more general case, we expand the previous eq. and

$$\begin{aligned}(n_l - n_u)(M + C) &= n_u P_{ul} - n_l P_{lu} \\ &= \frac{1}{2}[(n_u - n_l)(P_{lu} + P_{ul}) - (n_u + n_l)(P_{lu} - P_{ul})]\end{aligned}$$

Therefore, the population inversion is

$$\Delta n = \frac{\Delta n_0}{1 + \frac{2(M+C)}{P}} > 0 \quad \text{if} \quad \Delta n_0 > 0 \quad \text{i.e.,} \quad P_{lu} > P_{ul}$$



$$\frac{n_u}{n_l} \equiv \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right)$$

Then

$$T_{\text{ex}} = -\frac{T_0}{\ln\left(\frac{n_u g_l}{n_l g_u}\right)}$$

For maser, thus

$$T_{\text{ex}} < 0$$

Radiative Transfer of Maser

As mentioned before:

$$\frac{dI_\nu}{ds} = \frac{h\nu_0}{4\pi} [(n_u - n_l)BI_\nu + n_u A] \phi(\nu)$$

Since

$$M = B\bar{J} = BI \frac{\Omega_m}{4\pi}$$

where Ω_m is the beam solid angle of the radiation, we have

$$(n_u - n_l) = \Delta n = \frac{\Delta n_0}{1 + \frac{2C}{P} + \frac{2BI \frac{\Omega_m}{4\pi}}{P}}$$

Therefore,

$$\frac{dI_\nu}{ds} = \frac{\alpha_\nu I_\nu}{1 + I_\nu/I_s} + j_\nu$$

where

$$I_s = \frac{2\pi P}{B\Omega_m} \left(1 + \frac{2C}{P}\right)$$

$$\alpha_\nu = \frac{h\nu_0}{4\pi} B \frac{\Delta n_0}{1 + \frac{2C}{P}} \phi(\nu) \equiv \alpha_0 \phi(\nu)$$

$$j_\nu = \frac{h\nu_0}{4\pi} n_u A \phi(\nu) \equiv j_0 \phi(\nu)$$

Radiative Transfer of Maser

From previous slide,

$$\frac{dI_\nu}{ds} = \frac{\alpha_\nu I_\nu}{1 + I_\nu/I_s} + j_\nu$$

*** Let's focus on the **line center** at $\nu = \nu_0$ of the maser line. Two cases:

(1) **unsaturated maser, exponential amplification**: For $I_\nu \ll I_s$, we have

$$I_{\nu 0} = I_0 e^{\alpha_0 l} + \frac{j_0}{\alpha_0} (e^{\alpha_0 l} - 1)$$

where l is the length along the line of sight within the maser region and I_0 is background at ν_0 .

$$T_b = T_c e^{\alpha_0 l} + |T_{\text{ex}}| (e^{\alpha_0 l} - 1)$$

(2) **saturated maser, linear amplification**: For $I_\nu \gg I_s$, we have

$$\frac{dI_{\nu 0}}{ds} \approx \alpha_0 I_s + j_0 \quad \Rightarrow \quad I_{\nu 0} \approx I_0 + (\alpha_0 I_s + j_0)l$$

Line width of Unsaturated Maser

In most astrophysical applications the term j_ν is taken to be a constant and even can be dropped. Thus, for unsaturated maser, we can only consider the background amplification, ignoring the 2nd term. At any frequency, we have

$$T_b(\nu) = T_c e^{\alpha_\nu l}$$

Assuming a gaussian line profile then

$$\alpha_\nu = \alpha_0 \exp[-(\nu - \nu_0)^2 / 2\sigma_0^2]$$

Near the center of the line, we have

$$\alpha_\nu \approx \alpha_0 [1 - (\nu - \nu_0)^2 / 2\sigma_0^2]$$

Thus,

$$\begin{aligned} T_b(\nu) &= T_c \exp \{ \alpha_0 l [1 - (\nu - \nu_0)^2 / 2\sigma_0^2] \} \\ &= T_c \exp(\alpha_0 l) \exp \left[- \frac{\alpha_0 l (\nu - \nu_0)^2}{2\sigma_0^2} \right] \end{aligned}$$

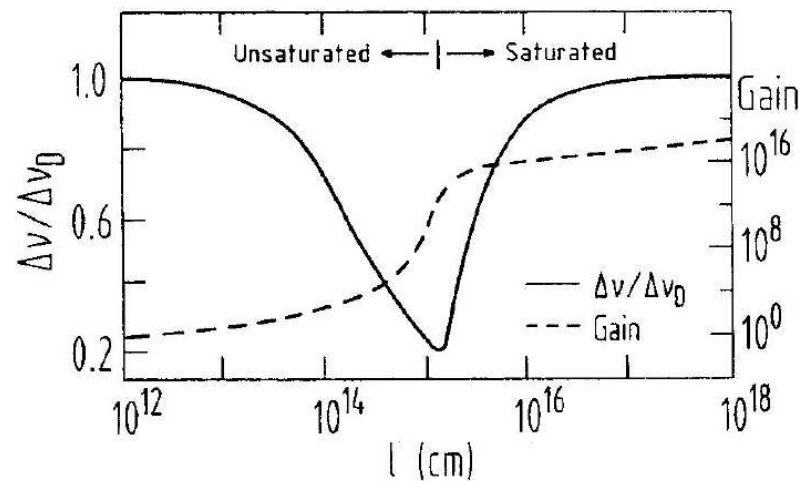
Therefore the dispersion and line width are

$$\sigma(\alpha_0 l) = \frac{\sigma_0}{\sqrt{\alpha_0 l}} \quad \text{and} \quad \text{FWHM} = \sqrt{8 \ln 2} \sigma$$

Unsaturated Maser

$$T_b(\nu) = T_c \exp(\alpha_0 l) \exp \left[- \frac{\alpha_0 l (\nu - \nu_0)^2}{2\sigma_0^2} \right]$$

$$\sigma(\alpha_0 l) = \frac{\sigma_0}{\sqrt{\alpha_0 l}} \quad \text{and} \quad \text{FWHM} = \sqrt{8 \ln 2} \sigma$$



The width will decrease until the center of the line begins to saturate. The linewidth then broadens as the wings of the line continue to experience exponential growth until the line again has the original line width σ_0 . In a saturated maser, the linewidth remains constant. The saturated and unsaturated masers are extreme cases. Intense masers are usually a mixture of these two categories: unsaturated when the signal intensity is small, but then saturated when the intensity becomes large. Note the exponential growth near the end of unsaturated case!

Masers

- physical condition required:
 - population inversion: still poorly understood
 - collisional pumping
 - radiative pumping

- physical information provided:
 - density/temperature (required for pumping)
 - accurate velocity (required for pumping)
 - kinematic information on “small” parcel of gas
 - accurate (relative) position information
 - proper motion measurements - distance

Masers

■ species

- OH, H₂O, SiO, H₂CO, CH₃OH, HCN, NH₃, H-recomb

■ locations

■ Galactic

- ISM: star forming region, HII regions

- OH, H₂O, SiO, H₂CO, CH₃OH, NH₃, H-recomb

- stellar: circumstellar envelopes around evolved stars

- OH, H₂O, SiO, HCN

- solar: comets

■ Extragalactic

- Water vapor megamaser ($\sim 10^6$ times brighter than galactic counterparts, or even gigamaser for OH)

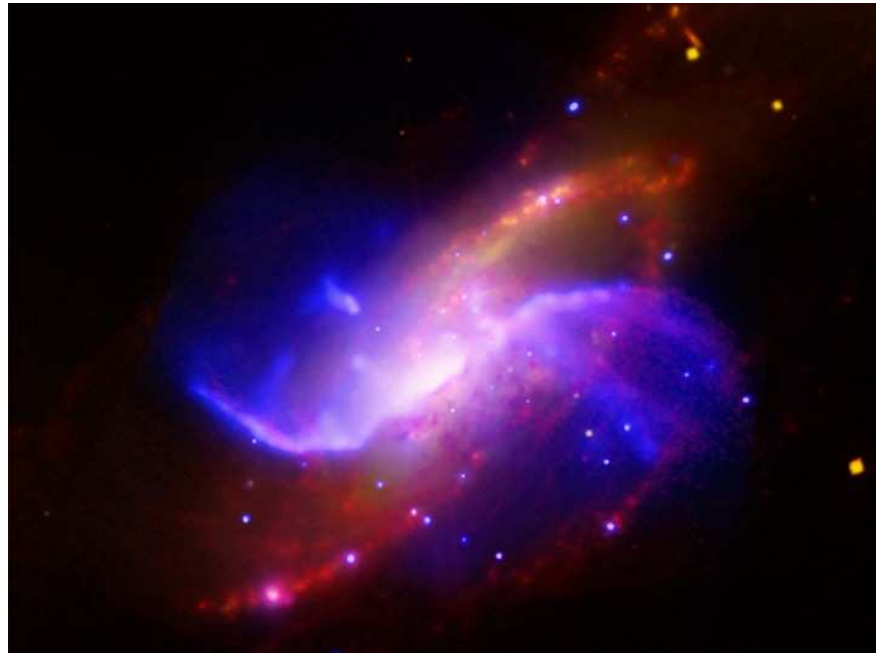
- H₂O in NGC 4945 (Santos & Lepine 1979)

- OH in Arp 220 (Baan 1982)

- circumnuclear disk

- amplification of radio continuum
- OH (FIR?), H₂CO (FIR/Radio?), H₂O (collisional?)
- star formation?

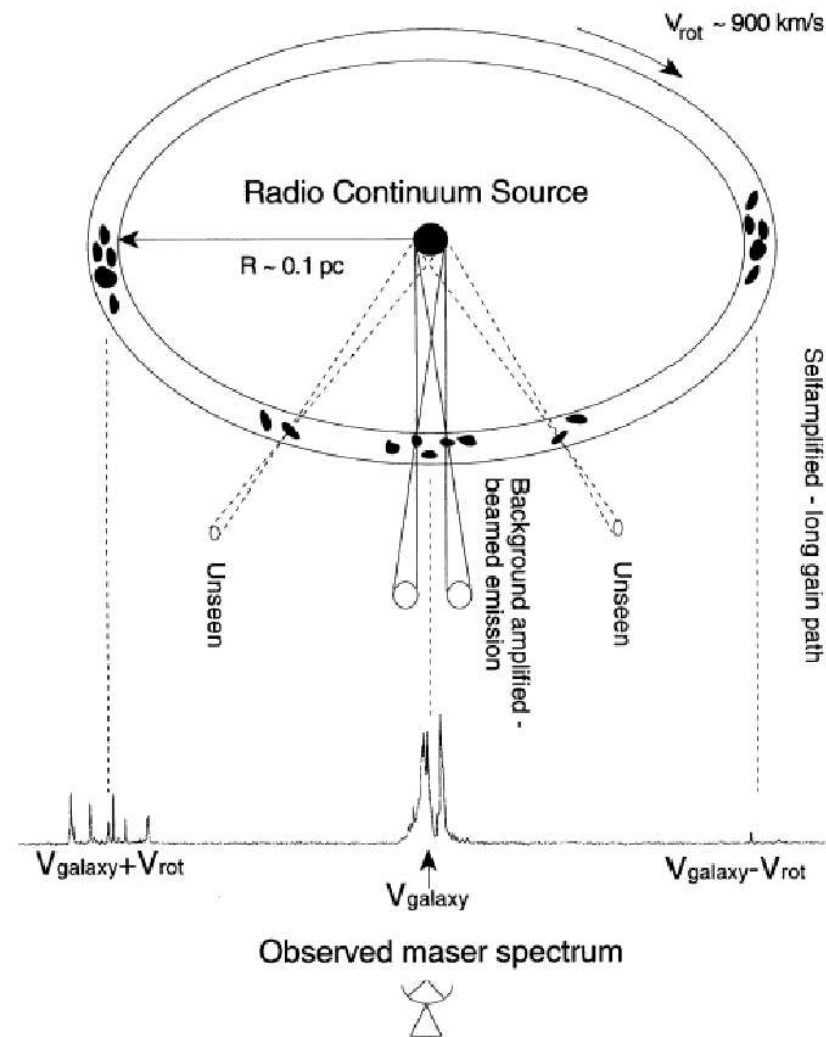
NGC 4259 or M106



Bright spiral galaxy NGC 4258 is about 30 thousand light years across and 21 million light years away toward the northern constellation Canes Venatici. The yellow and red hues in this composite image show the galaxy's sweeping spiral arms as seen in visible and infrared light. But x-ray and radio data (blue and purple) reveal two extra spiral arms – arms that don't align with the more familiar tracers of stars, gas, and dust. In fact, an analysis of the x-ray and radio data suggests that the anomalous arms are composed of material heated by shock waves. Detected at radio wavelengths, powerful jets originating in the galaxy's core likely drive the shocks into the disk of NGC 4258.

Megamasers

Megamasers toward SMBH in NGC 4258 (Seyfert-2 AGN)



Megamasers

H₂O Megamasers toward SMBH in NGC 4258 (Seyfert-2 AGN)

